

Quiz Advanced Functions

1. What is the purpose of the `map()` function in Python?
 - a) To filter elements from an iterable
 - b) To apply a function to each element of an iterable
 - c) To reduce an iterable to a single value
 - d) To sort elements of an iterable
2. Which of the following functions is NOT a part of the `functools` module?
 - a)
`map()`
 - b)
`filter()`
 - c)
`reduce()`
 - d)
`partial()`
3. What does the `filter()` function do?
 - a) Applies a function to each element of an iterable
 - b) Reduces an iterable to a single value
 - c) Filters elements from an iterable based on a condition
 - d) Sorts elements of an iterable
4. In Python, what is the purpose of the `reduce()` function?
 - a) To apply a function to each element of an iterable
 - b) To filter elements from an iterable
 - c) To concatenate strings or join lists
 - d) To apply a function to pairs of elements in an iterable until it's reduced to a single value

Coding exercises:

- Write a Python function `square_all(numbers)` that takes a list of numbers as input and returns a new list containing the square of each number in the input list. Use the `map()` function with a lambda function to implement this.
- Write a Python function `filter_positive(numbers)` that takes a list of numbers as input and returns a new list containing only the positive numbers from

the input list. Use the `filter()` function with a lambda function to implement this.

- Write a Python function `calculate_factorial(n)` that calculates the factorial of a given number `n`. Use the `reduce()` function with an appropriate lambda function to implement this.
- Write a Python function `count_vowels(string)` that takes a string as input and returns the count of vowels (a, e, i, o, u) in the input string. Use the `reduce()` function with an appropriate lambda function to implement this.